step t	0	1	2	3	4	5	6	7
slow	0	2	9	1	5	3	6	8
fast	0	9	5	6	7	1	3	8

meet at index 8
after 7 steps

fast jumps twice per step: `fast = nums[nums[fast]]`. Slow is inside the cycle at $t = 2$ (index 9), fast enters it at $t = 1$; from then on fast gains 1 position per step, so the gap closes in at most c steps.